

In [16]: `!nvidia-smi`

Mon Apr 27 17:19:21 2026

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+-----+
| NVIDIA-SMI 580.126.09      Driver Version: 580.126.09      CUDA Vers
ion: 13.0      |
+-----+-----+-----+-----+
+-----+
| GPU   Name                     Persistence-M | Bus-Id        Disp.A | Volatil
e Uncorr. ECC |
| Fan  Temp  Perf              Pwr:Usage/Cap |      Memory-Usage | GPU-Uti
l  Compute M. |
|               |              |
MIG M. |
|=====+=====+=====+=====+
|=====+
|   0   NVIDIA GeForce RTX 5090      Off |   00000000:01:00.0 Off |
N/A |
|  0%   35C    P8              12W /  600W |   579MiB /  32607MiB |      0%
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| Processes:
|
| GPU   GI    CI          PID    Type    Process name
GPU Memory |
|      ID    ID
Usage      |
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|=====+
|   0   N/A   N/A         1872     G    /usr/lib/xorg/Xorg
29MiB |
|   0   N/A   N/A         2036     G    /usr/bin/gnome-shell
15MiB |
|   0   N/A   N/A        136271    C    ...al_env_gpu_support/bin/python
500MiB |
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```

In [1]: `!pip install torch torchvision torchaudio --index-url https://download.pytor`

```

Looking in indexes: https://download.pytorch.org/whl/cu121
Collecting torch
  Downloading torch-2.5.1%2Bcu121-cp312-cp312-linux_x86_64.whl (780.4 MB)
    _____ 780.4/780.4 MB 3.5 MB/s 0:03:
370:00:0100:00
Collecting torchvision
  Downloading torchvision-0.20.1%2Bcu121-cp312-cp312-linux_x86_64.whl (7.3 M
B)
    _____ 7.3/7.3 MB 4.1 MB/s 0:00:01 e
ta 0:00:01
Collecting torchaudio
  Downloading torchaudio-2.5.1%2Bcu121-cp312-cp312-linux_x86_64.whl (3.4 MB)
    _____ 3.4/3.4 MB 4.7 MB/s 0:00:00 e
ta 0:00:01m
Collecting filelock (from torch)
  Downloading filelock-3.25.2-py3-none-any.whl.metadata (2.0 kB)
Requirement already satisfied: typing-extensions>=4.8.0 in /home/nazmul/glob
al_env_gpu_support/lib/python3.12/site-packages (from torch) (4.15.0)
Collecting networkx (from torch)
  Downloading networkx-3.6.1-py3-none-any.whl.metadata (6.8 kB)
Requirement already satisfied: jinja2 in /home/nazmul/global_env_gpu_suppor
t/lib/python3.12/site-packages (from torch) (3.1.6)
Collecting fsspec (from torch)
  Downloading fsspec-2026.2.0-py3-none-any.whl.metadata (10 kB)
Collecting nvidia-cuda-nvrtc-cu12==12.1.105 (from torch)
  Downloading nvidia_cuda_nvrtc_cu12-12.1.105-py3-none-manylinux1_x86_64.whl
(23.7 MB)
    _____ 23.7/23.7 MB 3.7 MB/s 0:00:06
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Collecting nvidia-cuda-runtime-cu12==12.1.105 (from torch)
  Downloading nvidia_cuda_runtime_cu12-12.1.105-py3-none-manylinux1_x86_64.w
hl (823 kB)
    _____ 823.6/823.6 kB 6.5 MB/s 0:00:
00eta 0:00:01
Collecting nvidia-cuda-cupti-cu12==12.1.105 (from torch)
  Downloading nvidia_cuda_cupti_cu12-12.1.105-py3-none-manylinux1_x86_64.whl
(14.1 MB)
    _____ 14.1/14.1 MB 3.5 MB/s 0:00:03
eta 0:00:01
Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch)
  Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl
(664.8 MB)
    _____ 664.8/664.8 MB 3.6 MB/s 0:03:
040:00:0100:05
Collecting nvidia-cublas-cu12==12.1.3.1 (from torch)
  Downloading nvidia_cublas_cu12-12.1.3.1-py3-none-manylinux1_x86_64.whl (41
0.6 MB)
    _____ 410.6/410.6 MB 3.6 MB/s 0:01:
530:00:0100:03
Collecting nvidia-cufft-cu12==11.0.2.54 (from torch)
  Downloading nvidia_cufft_cu12-11.0.2.54-py3-none-manylinux1_x86_64.whl (12
1.6 MB)
    _____ 121.6/121.6 MB 3.6 MB/s 0:00:
330:00:0100:01
Collecting nvidia-curand-cu12==10.3.2.106 (from torch)
  Downloading nvidia_curand_cu12-10.3.2.106-py3-none-manylinux1_x86_64.whl

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(56.5 MB)
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m0:00:0100:01
Collecting nvidia-cusolver-cu12==11.4.5.107 (from torch)
  Downloading nvidia_cusolver_cu12-11.4.5.107-py3-none-manylinux1_x86_64.whl
(124.2 MB)
_____ 124.2/124.2 MB 3.6 MB/s 0:00:
340:00:0100:02
Collecting nvidia-cuspars-cu12==12.1.0.106 (from torch)
  Downloading nvidia_cuspars-cu12-12.1.0.106-py3-none-manylinux1_x86_64.whl
(196.0 MB)
_____ 196.0/196.0 MB 3.6 MB/s 0:00:
540:00:0100:02
Collecting nvidia-nccl-cu12==2.21.5 (from torch)
  Downloading nvidia_nccl_cu12-2.21.5-py3-none-manylinux2014_x86_64.whl (18
8.7 MB)
_____ 188.7/188.7 MB 3.6 MB/s 0:00:
520:00:0100:02
Collecting nvidia-nvtx-cu12==12.1.105 (from torch)
  Downloading nvidia_nvtx_cu12-12.1.105-py3-none-manylinux1_x86_64.whl (99 k
B)
Collecting triton==3.1.0 (from torch)
  Downloading triton-3.1.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x
86_64.whl (209.6 MB)
_____ 209.6/209.6 MB 3.6 MB/s 0:00:
580:00:0100:02
Requirement already satisfied: setuptools in /home/nazmul/global_env_gpu_sup
port/lib/python3.12/site-packages (from torch) (82.0.1)
Collecting sympy==1.13.1 (from torch)
  Downloading sympy-1.13.1-py3-none-any.whl.metadata (12 kB)
Collecting nvidia-nvjitlink-cu12 (from nvidia-cusolver-cu12==11.4.5.107->tor
ch)
  Downloading nvidia_nvjitlink_cu12-12.9.86-py3-none-manylinux2010_x86_64.ma
nylinux_2_12_x86_64.whl.metadata (1.7 kB)
Collecting mpmath<1.4,>=1.1.0 (from sympy==1.13.1->torch)
  Downloading mpmath-1.3.0-py3-none-any.whl.metadata (8.6 kB)
Collecting numpy (from torchvision)
  Downloading numpy-2.4.3-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x
86_64.whl.metadata (6.6 kB)
Collecting pillow!=8.3.*,>=5.3.0 (from torchvision)
  Downloading pillow-12.1.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28
_x86_64.whl.metadata (8.8 kB)
Requirement already satisfied: MarkupSafe>=2.0 in /home/nazmul/global_env_gp
u_support/lib/python3.12/site-packages (from jinja2->torch) (3.0.3)
Downloading sympy-1.13.1-py3-none-any.whl (6.2 MB)
_____ 6.2/6.2 MB 3.9 MB/s 0:00:01 eta
0:00:01
Downloading mpmath-1.3.0-py3-none-any.whl (536 kB)
_____ 536.2/536.2 kB 2.8 MB/s 0:00:00
Downloading pillow-12.1.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x
86_64.whl (7.0 MB)
_____ 7.0/7.0 MB 3.6 MB/s 0:00:01 eta
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Downloading filelock-3.25.2-py3-none-any.whl (26 kB)
Downloading fsspec-2026.2.0-py3-none-any.whl (202 kB)
Downloading networkx-3.6.1-py3-none-any.whl (2.1 MB)
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2.1/2.1 MB 3.7 MB/s 0:00:00 eta
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Downloading numpy-2.4.3-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (16.6 MB)

16.6/16.6 MB 3.8 MB/s 0:00:04 e
ta 0:00:01
Downloading nvidia_nvjitlink_cu12-12.9.86-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl (39.7 MB)

39.7/39.7 MB 3.7 MB/s 0:00:10m
0:00:0100:01
Installing collected packages: mpmath, sympy, pillow, nvidia-nvtx-cu12, nvidia-nvjitlink-cu12, nvidia-nccl-cu12, nvidia-curand-cu12, nvidia-cufft-cu12, nvidia-cuda-runtime-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-cu12, numpy, networkx, fsspec, filelock, triton, nvidia-cuspars-cu12, nvidia-cudnn-cu12, nvidia-cusolver-cu12, torch, torchvision, torchaudio

23/23 [torchaudio]3 [torchaudio]
o]]ver-cu12]2]
Successfully installed filelock-3.25.2 fsspec-2026.2.0 mpmath-1.3.0 networkx-3.6.1 numpy-2.4.3 nvidia-cublas-cu12-12.1.3.1 nvidia-cuda-cupti-cu12-12.1.105 nvidia-cuda-nvrtc-cu12-12.1.105 nvidia-cuda-runtime-cu12-12.1.105 nvidia-cudnn-cu12-9.1.0.70 nvidia-cufft-cu12-11.0.2.54 nvidia-curand-cu12-10.3.2.106 nvidia-cusolver-cu12-11.4.5.107 nvidia-cuspars-cu12-12.1.0.106 nvidia-nccl-cu12-2.21.5 nvidia-nvjitlink-cu12-12.9.86 nvidia-nvtx-cu12-12.1.105 pillow-12.1.1 sympy-1.13.1 torch-2.5.1+cu121 torchaudio-2.5.1+cu121 torchvision-0.20.1+cu121 triton-3.1.0

```
In [15]: import torch
print(torch.cuda.is_available())
print(torch.cuda.get_device_name(0))
```

True
NVIDIA GeForce RTX 5090

```
In [3]: !pip install tensorflow
```

Collecting tensorflow
 Downloading tensorflow-2.21.0-cp312-cp312-manylinux_2_27_x86_64.whl.metadata (4.4 kB)

Collecting absl-py>=1.0.0 (from tensorflow)
 Downloading absl_py-2.4.0-py3-none-any.whl.metadata (3.3 kB)

Collecting astunparse>=1.6.0 (from tensorflow)
 Downloading astunparse-1.6.3-py2.py3-none-any.whl.metadata (4.4 kB)

Collecting flatbuffers>=25.9.23 (from tensorflow)
 Downloading flatbuffers-25.12.19-py2.py3-none-any.whl.metadata (1.0 kB)

Collecting gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 (from tensorflow)
 Downloading gast-0.7.0-py3-none-any.whl.metadata (1.5 kB)

Collecting google_pasta>=0.1.1 (from tensorflow)
 Downloading google_pasta-0.2.0-py3-none-any.whl.metadata (814 bytes)

Collecting libclang>=13.0.0 (from tensorflow)
 Downloading libclang-18.1.1-py2.py3-none-manylinux2010_x86_64.whl.metadata (5.2 kB)

Collecting opt_einsum>=2.3.2 (from tensorflow)
 Downloading opt_einsum-3.4.0-py3-none-any.whl.metadata (6.3 kB)

Requirement already satisfied: packaging in /home/nazmul/global_env_gpu_support/lib/python3.12/site-packages (from tensorflow) (26.2)

Collecting protobuf<8.0.0,>=6.31.1 (from tensorflow)
 Downloading protobuf-7.34.1-cp310-abi3-manylinux2014_x86_64.whl.metadata (595 bytes)

Requirement already satisfied: requests<3,>=2.21.0 in /home/nazmul/global_env_gpu_support/lib/python3.12/site-packages (from tensorflow) (2.33.1)

Requirement already satisfied: setuptools in /home/nazmul/global_env_gpu_support/lib/python3.12/site-packages (from tensorflow) (82.0.1)

Requirement already satisfied: six>=1.12.0 in /home/nazmul/global_env_gpu_support/lib/python3.12/site-packages (from tensorflow) (1.17.0)

Collecting termcolor>=1.1.0 (from tensorflow)
 Downloading termcolor-3.3.0-py3-none-any.whl.metadata (6.5 kB)

Requirement already satisfied: typing_extensions>=3.6.6 in /home/nazmul/global_env_gpu_support/lib/python3.12/site-packages (from tensorflow) (4.15.0)

Collecting wrapt>=1.11.0 (from tensorflow)
 Downloading wrapt-2.1.2-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (7.4 kB)

Collecting grpcio<2.0,>=1.24.3 (from tensorflow)
 Downloading grpcio-1.80.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (3.8 kB)

Collecting keras>=3.12.0 (from tensorflow)
 Downloading keras-3.14.0-py3-none-any.whl.metadata (6.3 kB)

Requirement already satisfied: numpy>=1.26.0 in /home/nazmul/global_env_gpu_support/lib/python3.12/site-packages (from tensorflow) (2.4.3)

Collecting h5py<3.15.0,>=3.11.0 (from tensorflow)
 Downloading h5py-3.14.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (2.7 kB)

Collecting ml_dtypes<1.0.0,>=0.5.1 (from tensorflow)
 Downloading ml_dtypes-0.5.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (8.9 kB)

Requirement already satisfied: charset_normalizer<4,>=2 in /home/nazmul/global_env_gpu_support/lib/python3.12/site-packages (from requests<3,>=2.21.0->tensorflow) (3.4.7)

Requirement already satisfied: idna<4,>=2.5 in /home/nazmul/global_env_gpu_support/lib/python3.12/site-packages (from requests<3,>=2.21.0->tensorflow) (3.13)

Requirement already satisfied: urllib3<3,>=1.26 in /home/nazmul/global_env_g

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pu_support/lib/python3.12/site-packages (from requests<3,>=2.21.0->tensorflo
w) (2.6.3)
Requirement already satisfied: certifi>=2023.5.7 in /home/nazmul/global_env_
gpu_support/lib/python3.12/site-packages (from requests<3,>=2.21.0->tensorfl
ow) (2026.4.22)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /home/nazmul/global_env_
gpu_support/lib/python3.12/site-packages (from astunparse>=1.6.0->tensorflo
w) (0.47.0)
Collecting rich (from keras>=3.12.0->tensorflow)
  Downloading rich-15.0.0-py3-none-any.whl.metadata (18 kB)
Collecting namex (from keras>=3.12.0->tensorflow)
  Downloading namex-0.1.0-py3-none-any.whl.metadata (322 bytes)
Collecting optree (from keras>=3.12.0->tensorflow)
  Downloading optree-0.19.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28
_x86_64.whl.metadata (34 kB)
Collecting markdown-it-py>=2.2.0 (from rich->keras>=3.12.0->tensorflow)
  Downloading markdown_it_py-4.0.0-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /home/nazmul/globa
l_env_gpu_support/lib/python3.12/site-packages (from rich->keras>=3.12.0->te
nsorflow) (2.20.0)
Collecting mdurl~=0.1 (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tens
orflow)
  Downloading mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Downloading tensorflow-2.21.0-cp312-cp312-manylinux_2_27_x86_64.whl (572.6 M
B)
_____ 572.6/572.6 MB 3.6 MB/s 0:02:39
m0:00:0100:05
Downloading grpcio-1.80.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x8
6_64.whl (6.8 MB)
_____ 6.8/6.8 MB 4.2 MB/s 0:00:01 eta
0:00:01
Downloading h5py-3.14.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_
64.whl (4.9 MB)
_____ 4.9/4.9 MB 3.6 MB/s 0:00:01 eta
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Downloading ml_dtypes-0.5.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28
_x86_64.whl (5.0 MB)
_____ 5.0/5.0 MB 3.9 MB/s 0:00:01 eta
0:00:01
Downloading protobuf-7.34.1-cp310-abi3-manylinux2014_x86_64.whl (324 kB)
Downloading absl_py-2.4.0-py3-none-any.whl (135 kB)
Downloading astunparse-1.6.3-py2.py3-none-any.whl (12 kB)
Downloading flatbuffers-25.12.19-py2.py3-none-any.whl (26 kB)
Downloading gast-0.7.0-py3-none-any.whl (22 kB)
Downloading google_pasta-0.2.0-py3-none-any.whl (57 kB)
Downloading keras-3.14.0-py3-none-any.whl (1.6 MB)
_____ 1.6/1.6 MB 4.1 MB/s 0:00:00 eta
0:00:01
Downloading libclang-18.1.1-py2.py3-none-manylinux2010_x86_64.whl (24.5 MB)
_____ 24.5/24.5 MB 3.7 MB/s 0:00:06m
0:00:0100:01
Downloading opt_einsum-3.4.0-py3-none-any.whl (71 kB)
Downloading termcolor-3.3.0-py3-none-any.whl (7.7 kB)
Downloading wrapt-2.1.2-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.
manylinux_2_5_x86_64.whl (121 kB)
Downloading namex-0.1.0-py3-none-any.whl (5.9 kB)
```

Downloading optree-0.19.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (451 kB)
Downloading rich-15.0.0-py3-none-any.whl (310 kB)
Downloading markdown_it_py-4.0.0-py3-none-any.whl (87 kB)
Downloading mdurl-0.1.2-py3-none-any.whl (10.0 kB)
Installing collected packages: namex, libclang, flatbuffers, wrapt, termcolor, protobuf, optree, opt_einsum, ml_dtypes, mdurl, h5py, grpcio, google_pasta, gast, absl-py, markdown-it-py, astunparse, rich, keras, tensorflow

20/20 [tensorflow]0 [tensorflow]
Successfully installed absl-py-2.4.0 astunparse-1.6.3 flatbuffers-25.12.19 gast-0.7.0 google_pasta-0.2.0 grpcio-1.80.0 h5py-3.14.0 keras-3.14.0 libclang-18.1.1 markdown-it-py-4.0.0 mdurl-0.1.2 ml_dtypes-0.5.4 namex-0.1.0 opt_einsum-3.4.0 optree-0.19.0 protobuf-7.34.1 rich-15.0.0 tensorflow-2.21.0 termcolor-3.3.0 wrapt-2.1.2

```
In [6]: import tensorflow as tf
        print(tf.config.list_physical_devices('GPU'))
```

```
[PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
```

```
In [14]: import tensorflow as tf

        gpus = tf.config.list_physical_devices('GPU')

        if gpus:
            for gpu in gpus:
                details = tf.config.experimental.get_device_details(gpu)
                print(details.get('device_name', 'Unknown GPU'))
        else:
            print("No GPU found")
```

NVIDIA GeForce RTX 5090